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AI/ML Intern

Key Learnings Report

Introduction

During my internship at RARITONE, I gained practical exposure to Artificial Intelligence, Machine Learning, Computer Vision, and Virtual Try-On technologies. The internship involved researching modern virtual fitting solutions, studying industry-standard datasets and models, and understanding how multiple technologies work together to create intelligent fashion applications.

Technical Learnings

1. Computer Vision Fundamentals

Developed an understanding of how computer vision techniques are used to analyze human images and extract meaningful information for virtual try-on applications.

Topics Learned:

- Image Processing
- Human Segmentation
- Feature Extraction
- Image Preprocessing

2. Pose Estimation and Landmark Detection

Learned how AI models detect body landmarks and estimate human poses using computer vision techniques.

Key Concepts:

- Body Keypoints
- Pose Detection
- Landmark Extraction
- Human Body Analysis

Tools Explored:

- MediaPipe Pose
- OpenCV

3. Body Measurement Estimation

Studied methods for estimating body measurements from images using body landmarks and proportional analysis.

Measurements Analyzed:

- Height
- Shoulder Width
- Waist Position
- Hip Position
- Body Proportions

4. Virtual Try-On Systems

Developed an understanding of how virtual try-on systems combine segmentation, pose estimation, garment alignment, and image generation.

Areas Explored:

- 2D Virtual Try-On
- 3D Virtual Try-On
- Garment Processing
- Avatar-Based Visualization

5. AI Models and Frameworks

Researched various tools, frameworks, and models used in virtual try-on applications.

Technologies Studied:

- MediaPipe
- OpenCV
- NumPy
- Trimesh
- Pyrender
- Unity
- Unreal Engine

6. Dataset Analysis

Learned the importance of high-quality datasets in training and evaluating AI systems.

Datasets Explored:

- DeepFashion
- DeepFashion2
- VITON Dataset
- Dress Code Dataset
- Fashionpedia

Professional Learnings

Research Skills

- Conducting technical research
- Comparing models and frameworks
- Evaluating datasets
- Analyzing emerging technologies

Documentation Skills

- Preparing technical reports
- Creating workflow diagrams
- Writing research summaries
- Presenting findings professionally

Problem Solving

- Resolving environment setup issues
- Managing library dependencies
- Debugging implementation challenges
- Exploring alternative solutions

Key Takeaways

- AI-powered virtual try-on systems require the integration of multiple technologies rather than a single model.
- Human segmentation and pose estimation are critical components of virtual fitting solutions.
- Accurate body measurements significantly improve recommendation quality.
- 3D virtual try-on systems offer greater realism but involve higher complexity.
- Proper documentation and workflow design are essential for successful AI projects.

Conclusion

The internship provided valuable hands-on learning in Artificial Intelligence, Machine Learning, and Computer Vision. It strengthened my technical knowledge, research abilities, and problem-solving skills while giving me practical insight into the development of AI-powered virtual try-on systems. The experience has established a strong foundation for future work in AI, computer vision, and intelligent recommendation systems.